MIL-S-63171(AR)
25 June 1982
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

SEAL, BALL, MONITORING PORT
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for monitoring port ball seals used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-G-45204	Gold Plating, Electrodeposited

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

11829182	Seal, Ball, Monitoring Port
11829184	Process for Gold Plating

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B-567	Measurement of Coating Thickness by the Beta Backscatter Method

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Construction. The monitoring port ball seal shall comply with all requirements specified on drawing (dwg.) 11829182, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

3.3 Plating. The monitoring port ball seal shall be nickel and gold plated in accordance with dwg. 11829184.

3.3.1 Plating quality. Gold plating shall be of high quality, fine-grained and free of nodules, bare spots, scratches, flaking, blistering or any other obvious plating defects. Testing shall be performed in accordance with 4.5.2.1.

3.3.2 Plating thickness. Gold plating thickness shall be as specified on the dwg. Testing shall be performed in accordance with 4.5.2.2.

3.3.3 Adhesion. Gold plating shall exhibit no evidence of defective adhesion such as separation from the base material. Testing shall be performed in accordance with 4.5.2.3.

3.4 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.5 Workmanship. The monitoring port ball seal shall be fabricated and finished in a thoroughly workmanlike manner and shall be free of scratches, nicks, dents, burrs, or any other obvious physical defects. Parts shall be clean and free of dirt, grease or any other foreign matter. Cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified

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herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.4), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Monitoring port ball seal (plated)	11829182	6
Monitoring port ball seal (unplated)	11829182	6

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspection specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63171(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Seal, Ball, Monitoring Port	1 OF 1	11829182
			NEXT HIGHER ASSEMBLY
			M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of defects (unplated)	6			4.4.2.1
	Outside diameter				
	*Hardness				
	Classification of defects (plated)	6			4.4.2.1
	Surface finish and gold plating quality				
	*Gold plating thickness and adhesion				
	Workmanship				
	Material				

NOTES:

*Tests preceded by an asterisk are destructive and shall be performed last.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of monitoring port ball seals shall comply with the lot formation provisions of MIL-A-48078. In addition a plating inspection lot shall consist of items plated at the same time with the same equipment by the same plating process with the same lot of plating material. A plating inspection lot shall not exceed 1000 units.

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Tables herein.

(a) Major defects. Unless otherwise specified, inspection for Major defects shall be sampled in accordance with Table I.

(b) Minor defects. Unless otherwise specified, inspection for Minor defects shall be sampled in accordance with Table II.

Table I. Sampling plan for major defects.

LQ = .05 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table II. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63171(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Seal, Ball, Monitoring Port	1 OF 1	11829182
			NEXT HIGHER ASSEMBLY: M753 Proj

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter (unplated)		Table I	3.1	Gage
*102.	Hardness (unplated)		Table I	3.1	4.5.1
103.	Nickel plating present (before gold plating)		100%	3.1	Visual
104.	Gold plating quality		100%	3.3.1	4.5.2.1
105.	Gold plating thickness (specimens)	4.4.3 1		3.3.2	4.5.2.2
106.	Gold plating adhesion (specimens)	4.4.3 1		3.3.3	4.5.2.3
107.	Surface finish		100%	3.1	Visual
108.	Materials		----	3.2	Visual
Minor:					
201.	Poor workmanship		Table II	3.5	Visual

NOTES:

*These tests are considered destructive to this item. Do not return tested items to the inspection lot.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63171(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Packaging				See 5.0
					NEXT HIGHER ASSEMBLY
					N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		---	5.0	Visual
102.	Objective evidence (see 3.2) not available for packaging and packing material		---	5.1.1	Visual
103.	Parts not clean and free of dirt and other foreign material prior to packaging		100%	5.1.1	Visual
104.	Parts not individually packaged		100%	5.1.1	Visual
105.	Stiffening and cushioning of packages inadequate to provide firm package (when required)		100%	5.1.1	Visual/Manual
106.	Individual packages not properly heat sealed		100%	5.1.1	Visual
107.	Intermediate pack improper or not properly sealed		100%	5.1.2	Visual/Manual
108.	Exterior pack improper or not properly sealed		100%	5.1.3	Visual
109.	Marking of individual packages, intermediate packs or exterior packs missing, improper or illegible		100%	5.1.4	Visual
110.	Evidence of poor workmanship		100%	-----	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and herein.

4.4.3.1 Plating thickness and adhesion. Specimens or coupons from each gold plating inspection lot plated concurrently with the actual items shall be tested for plating thickness and adhesion in accordance with 4.5.2. The specimens or coupons shall be prepared from the same base material as the actual items, with the same finish, and cleaned and processed concurrently with the items being plated (see dwg. 11829184). The thickness and adhesion shall be verified on at least two specimens or coupons per plating inspection lot. If any specimen or coupon fails to meet the applicable thickness or adhesion requirements, the plating inspection lot under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods.

4.5.1 Hardness. Rockwell hardness shall be measured in accordance with ASTM E-18 or an approved equivalent.

4.5.2 Plating (see 3.3).

4.5.2.1 Plating quality. Gold plating quality shall be examined on each item under 10x magnification.

4.5.2.2 Plating thickness. Gold plating thickness shall be measured in accordance with ASTM B-567 or other approved test method.

4.5.2.3 Adhesion. Gold plating adhesion shall be examined in accordance with 4.5.2.2 of MIL-G-45204 by shearing the plating down to the base metal and examining the base metal/plating bond for adhesion under 4x magnification.

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5. PACKAGING.

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be single item packaged in fiberboard containers, sealed in waterproof barrier bags and heat sealed in accordance with MIL-STD-1169, Level A. Barrier bags shall be fabricated from MIL-B-121 (Type II, Grade A, Class 1) material or an approved equivalent. Cushioning adequate to prevent physical damage to the item and to finished surfaces during transit, storage and handling shall be provided as required.

5.1.2 Intermediate packing. Single item packages shall be packed in intermediate packages in accordance with MIL-STD-1169, Level B.

5.1.3 Packing. Intermediate packages shall be packed in accordance with MIL-STD-1169, Level B.

5.1.4 Marking. Unit packages, intermediate packs and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no
c. Used on M753 Projectile
d. Manufacturer
e. Lot no.
f. Contract or purchase order no.
g. Date of acceptance

6. NOTES

6.1 Intended use. Monitoring port ball seals covered by this specification are intended for use on rocket motor/bulkhead assemblies for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.4).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification) supported by test data that each lot of material meets the requirements of the applicable dwgs. and specifications.

6.5 Supersession data. This specification includes the requirements of AR-PD-18 dated 9 April, 1979.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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Preparing activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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